Physics Olympiad L1-17

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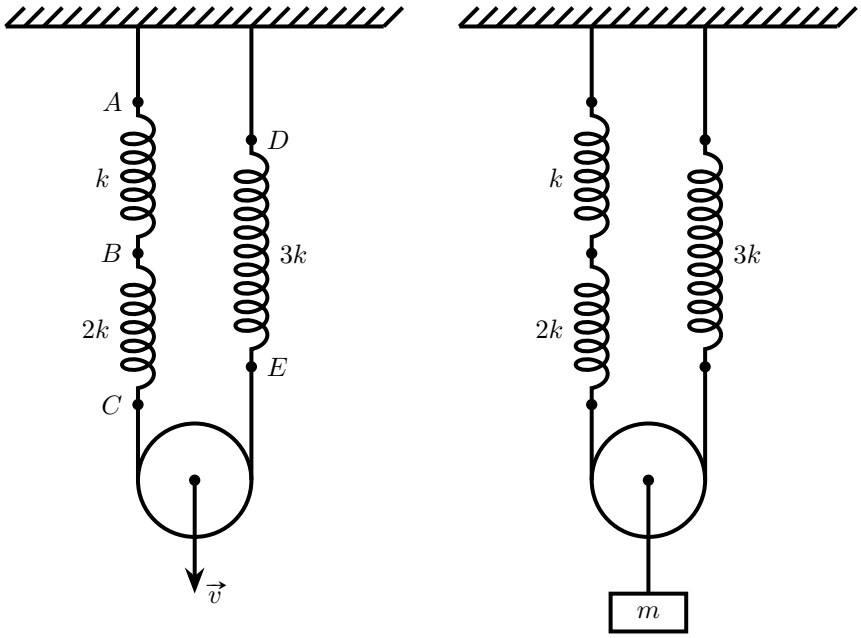
Problem 1: Spring-Pulley System

Three massless springs with spring constants k, $2k$ and $3k$ are connected by inextensible strings to a horizontal ceiling and a massless pulley as shown. At a certain moment, under the action of an external force $\vec{F}$ that increases smoothly with time, the pulley starts moving downward with a constant velocity $\vec{v}$.

- Find the velocities of points A, B, C, D and E;
- Find the velocity of the string CE relative to the rim of the pulley;
- Find the time dependence of the force $\vec{F}(t)$ if it is known that $\vec{F}(0) = \vec{0}$;

Now assume that the Spring-Pulley system is at rest at height $y = 0$, where the y-axis points downwards. At $t = 0$, a body of mass m is attached to the pulley and released from rest.

- Find the equation of motion $y(t)$ of the body;
- Find the period T and angular frequency ω of the oscillations;
- Find the maximum depth d that the body reaches during this motion;



Problem 2: Metal Ring in Magnetic Field

A metal ring of radius $r = 0{,}1\,\text{m}$ and resistance $R = 2{,}0\,\Omega$ is positioned horizontally in a uniform vertical magnetic field of magnitude $B_0 = 5{,}0 \cdot 10^{-5}\,\text{T}$. It is then flipped over so that its other side faces upward.

- What magnitude of electric charge q passes through the cross section of the ring?
- The ring is flipped uniformly in a time $\tau = 1\,\text{s}$. Find the magnitude of the current as a function of time;

Thermal effects and self-inductance can be ignored.

Problem 3: Counting Waves

Between two points on a sound wave, which vibrate in phase, there are $N = 825$ wavelengths. When the temperature of the medium is increased by $\Delta T_0 = 1\,\text{K}$, the speed of sound increases by $\eta = 0{,}2\,\%$.

- Find the temperature T of the medium;
- What temperature increase ΔT is required for the two points to start vibrating in antiphase?

The medium is assumed to be an ideal gas and the frequency of the sound remains constant.

Problem 4: **Resultant of Parallel Forces**

Two parallel and equally directed forces F_1 and F_2 are applied to a rigid body. Prove that:

- The resultant force is parallel to the applied forces and directed in the same direction;
- The magnitude of the resultant force is equal to the sum of the magnitudes of the applied forces;
- The line of action of the resultant passes through a point that divides the segment between the points of application of the two forces into parts that are inversely proportional to the force magnitudes;

Problem 5: **Electron-Ring Interactions**

We consider the motion of an electron along the axis of a uniformly charged ring with linear charge density $\lambda > 0$. Assuming the mass of the ring to be much larger than that of the electron, find:

- With what speed will the e^- pass through the centre of the ring if it is attracted from infinity?
- With what speed will the e^- pass through the centre of the ring if it is attracted from distance ηR?
- What is the period of small oscillations of the e^- about the centre of the ring along its axis?
- What is the largest amplitude for which the magnitude of the restoring force differs from that predicted by the small-oscillation approximation by no more than 1 %? Specify your answer as a multiple of R.
- What is the largest amplitude for which the period of oscillation differs from that predicted by the small-oscillation approximation by no more than 1 %? Specify your answer as a multiple of R.

The electron has mass m_e and charge $-e$. The radius of the ring is R and η is a real number ($\eta \in \mathbb{R}^+$). In all situations where relevant, the electron starts its motion from rest. You may use appropriate approximations and/or cutoffs where necessary.

Problem 6: **Free Expansion**

One of the two thermally insulated vessels interconnected by a tube with valve contains $\nu = 2{,}2$ moles of an ideal gas. The other vessel is evacuated. The valve having been opened, the gas increases its volume $n = 3{,}0$ times. Find the entropy increment of the gas.

Appendix

Useful Mathematical Formulae and Physical Constants

Important: Infinite series that are marked with $(*)$ are only valid under the condition that $|x| < 1$.

Taylor Expansions

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} + \mathcal{O}(x^7)$$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \mathcal{O}(x^8)$$

$$\mathrm{tg}(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + \mathcal{O}(x^7)$$

$$\arcsin(x) = x + \frac{x^3}{6} + \frac{3x^5}{40} + \mathcal{O}(x^7)$$

$$\mathrm{arctg}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} + \mathcal{O}(x^7)$$

$$\exp(x) = 1 + x + \frac{x^2}{2} + \mathcal{O}(x^3)$$

$$\ln(1+x) = x - \frac{1}{2}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1-x)^{-1} = 1 + x + x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{-1} = 1 - x + x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{+\frac{1}{2}} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{-\frac{1}{2}} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{+\frac{3}{2}} = 1 + \frac{3}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{-\frac{3}{2}} = 1 - \frac{3}{2}x + \frac{15}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

Inverse Hyperbolic Functions

$$\mathrm{arsinh}(x) = \ln(x + \sqrt{x^2+1})$$

$$\mathrm{arcosh}(x) = \ln(x + \sqrt{x^2-1})$$

$$\mathrm{artgh}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

$$\mathrm{arctgh}(x) = \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$$

Indefinite Integrals

$$\int (1-x^2)^{-\frac{3}{2}}\,dx = \frac{x}{\sqrt{1-x^2}} + C$$

$$\int (1+x^2)^{-\frac{3}{2}}\,dx = \frac{x}{\sqrt{1+x^2}} + C$$

$$\int (1-x^2)^{-\frac{1}{2}}\,dx = \arcsin(x) + C$$

$$\int (1+x^2)^{-\frac{1}{2}}\,dx = \mathrm{arsinh}(x) + C$$

$$\int (1-x^2)^{+\frac{1}{2}}\,dx = \frac{\arcsin(x) + x\sqrt{1-x^2}}{2} + C$$

$$\int (1+x^2)^{+\frac{1}{2}}\,dx = \frac{\mathrm{arsinh}(x) + x\sqrt{1+x^2}}{2} + C$$

Useful Physical Constants

Constant	Symbol	Value	Measuring Units
Light speed in vacuum	c	$3{,}000 \times 10^{8}$	m/s
Rest mass of electron	m_e	$9{,}109 \times 10^{-31}$	kg
Rest mass of proton	m_p	$1{,}673 \times 10^{-27}$	kg
Rest mass of neutron	m_n	$1{,}675 \times 10^{-27}$	kg
Rest mass of muon	m_μ	$1{,}884 \times 10^{-28}$	kg
Rest mass of tau-lepton	m_τ	$3{,}168 \times 10^{-27}$	kg
Elementary charge	e	$1{,}602 \times 10^{-19}$	C
Planck's constant	h	$6{,}626 \times 10^{-34}$	Js
Boltzmann's constant	k	$1{,}381 \times 10^{-23}$	J/K
Gravitational constant	γ	$6{,}673 \times 10^{-11}$	$\mathrm{Nm^2/kg^2}$
Dielectric permittivity of vacuum	ε_0	$8{,}854 \times 10^{-12}$	F/m
Avogadro's number	N_A	$6{,}022 \times 10^{23}$	$\mathrm{mol^{-1}}$
Rydberg's constant	R_∞	$1{,}097 \times 10^{7}$	$\mathrm{m^{-1}}$
Magnetic permeability of vacuum	μ_0	$4\pi \times 10^{-7}$	H/m
Universal gas constant	R	8,314	$\mathrm{J/(mol \cdot K)}$